

The Constrained Optimisation Problem

EC306: Intermediate Microeconomics — Day 2

Core reading: Perloff (2020), Ch. 4

Recap and today's move

- **Yesterday:** what the consumer *wants* — preferences, MRS, utility as an ordinal summary.
- **Today:** what the consumer *can afford*, and how the two collide to produce a single chosen bundle — and ultimately, a demand function.

Roadmap

- ➊ The budget set and budget line
- ➋ The tangency condition (intuition first)
- ➌ The Lagrangian: a recipe, and what λ *means*
- ➍ Interior solutions: Cobb–Douglas and the expenditure-share result
- ➎ Corner solutions: when tangency fails
- ➏ Policy application: cash vs in-kind transfers

The budget line: prices convert goods into money

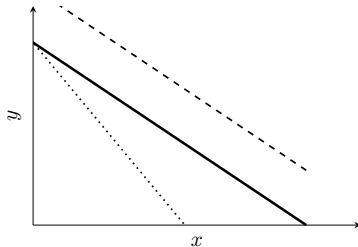
With income m and prices p_x, p_y , affordable bundles satisfy $p_x x + p_y y \leq m$. The boundary — the **budget line** — is

$$y = \frac{m}{p_y} - \frac{p_x}{p_y} x, \quad \text{slope} = -\frac{p_x}{p_y}.$$

Read the slope

$-p_x/p_y$ is the *market* rate of exchange: how much y the market forces you to give up for one more x . Compare this to the MRS, which is the rate you'd *willingly* give up. The optimum is where the two agree.

What moves the budget line

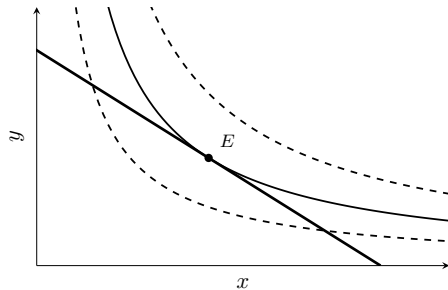


Solid: baseline. **Dashed:** income $m \uparrow$ — parallel outward shift. **Dotted:** $p_x \uparrow$ — pivots inward about the y -intercept (only x got dearer).

Why pivots, not shifts, matter

A pure price change *rotates* the line. This is the seed of the substitution effect we unpack tomorrow: relative prices, not just purchasing power, have changed.

The optimum, before any calculus



The best affordable bundle E is where the highest reachable indifference curve just *touches* the budget line. Lower curve (inner dashed): affordable but wasteful. Higher curve (outer dashed): preferred but unaffordable.

The tangency condition

At an interior optimum the indifference curve and budget line share a slope:

$$\underbrace{\text{MRS} = \frac{\text{MU}_x}{\text{MU}_y}}_{\text{willing trade-off}} = \underbrace{\frac{p_x}{p_y}}_{\text{market trade-off}} .$$

Rearrange into the **equal-bang-for-buck** form:

$$\boxed{\frac{\text{MU}_x}{p_x} = \frac{\text{MU}_y}{p_y}}$$

Intuition

The last pound spent on streaming must buy the same extra utility as the last pound spent on coffee. If not, reallocate a pound toward the higher-return good and gain. At the optimum, no such gain remains.

The Lagrangian: a mechanical recipe

Form

$$\mathcal{L}(x, y, \lambda) = u(x, y) + \lambda(m - p_x x - p_y y).$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = m - p_x x - p_y y = 0.$$

Dividing the first two recovers the tangency condition $MU_x/MU_y = p_x/p_y$. The third is just the budget, holding with equality (by monotonicity, you spend it all).

What is λ ? The marginal utility of income

From the FOCs, $\lambda = MU_x/p_x = MU_y/p_y$. So λ is the common “bang per buck”.

The envelope reading

$\lambda = \frac{\partial u^*}{\partial m}$: if income rose by one pound, utility rises by λ . It is the **shadow price** of the budget constraint — the value of relaxing it by a unit.

Where you meet λ in practice

Shadow prices are the entire logic of cost-benefit analysis and shadow pricing in public projects: “what is one more unit of the binding resource worth?” Same object, different label.

Worked example — Cobb–Douglas demand

$u = x^a y^b$, budget $p_x x + p_y y = m$. Tangency:

$$\frac{\text{MU}_x}{\text{MU}_y} = \frac{ax^{a-1}y^b}{bx^ay^{b-1}} = \frac{ay}{bx} = \frac{p_x}{p_y} \Rightarrow p_y y = \frac{b}{a} p_x x.$$

Substitute into the budget $p_x x + \frac{b}{a} p_x x = m$:

$$x^* = \frac{a}{a+b} \frac{m}{p_x}, \quad y^* = \frac{b}{a+b} \frac{m}{p_y}.$$

The expenditure-share result

Spending on x is $p_x x^* = \frac{a}{a+b} m$: a *constant fraction* of income, independent of prices. This is why Cobb–Douglas is the empirical workhorse — household budget shares across broad categories really are roughly stable.

Your turn (1) — in-lecture

Your turn

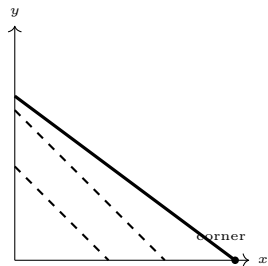
$u = x^{0.25}y^{0.75}$, $m = 120$, $p_x = 2$, $p_y = 3$. Find x^* , y^* , and the share of income spent on each good.

When tangency fails: corner solutions

Tangency assumes an *interior* optimum with a smooth IC. Two cases break it:

Perfect substitutes $u = ax + by$. Constant $MRS = a/b$. If $a/b > p_x/p_y$, x is the better deal at *every* point — spend everything on x :

$$x^* = m/p_x, \quad y^* = 0.$$



Steeper IC than budget $\Rightarrow$ corner on the x -axis.

Real world

“All-or-nothing” choices: a buyer who treats two suppliers’ bulk commodity as identical buys 100% from whoever is fractionally cheaper. Switching is total, not gradual.

Quasilinear: a partial corner story

$u = v(x) + y$ with v concave, e.g. $u = 8\sqrt{x} + y$. FOC on x : $v'(x) = p_x/p_y \Rightarrow x^*$ solves $\frac{4}{\sqrt{x}} = \frac{p_x}{p_y}$,
independent of m .

The income kink

Above a threshold income, all extra income goes to y ; x^* is fixed by prices alone — no income effect on x . Below it, you hit a corner with $y = 0$. This is the standard model for a “small” good whose consumption doesn’t track your salary.

Policy: why economists prefer cash to in-kind transfers

A household at optimum E_0 . Government can give either (i) $\pounds T$ cash, or (ii) $\pounds T$ of food vouchers (food only).

- **Cash:** parallel outward shift; household re-optimises freely; reaches the highest curve attainable from the new budget.
- **Voucher:** budget shifts out only in the food direction; if the household *wanted* to spend less than T on food, it is now constrained — it lands on a (weakly) *lower* indifference curve than cash.

The result, and its limit

For any household whose desired food spending exceeds T , the voucher is *equivalent* to cash. The welfare loss only bites for “inframarginal” households. This is the precise efficiency case — but note it ignores paternalism, intra-household bargaining, and externalities, which is why in-kind transfers persist. Say so in the exam.

Your turn (2) — in-lecture

Your turn

A consumer with $u = \min\{x, y\}$, $m = 60$, $p_x = 2$, $p_y = 4$. Find the optimal bundle. (Hint: tangency is useless here — use the kink condition.)

What you can now do

- ① Draw a budget set and predict how it shifts under income and price changes.
- ② Solve the consumer's problem by tangency and by Lagrangian, and interpret λ .
- ③ Derive demand functions and the expenditure-share result for Cobb–Douglas.
- ④ Recognise and solve corner solutions where tangency fails.
- ⑤ Use indifference-curve geometry to evaluate a real transfer policy.

Tomorrow

We let prices and income *vary* and trace out the **demand curve** — then decompose any price change into substitution and income effects, and measure responsiveness with elasticities.